Projective Algebraic Stability and Exact Degree Doubling for a Quadratic Hénon Automorphism

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Abstract

We freeze the polynomial automorphism $H(x,y) = (x^2 - 4 - y, x)$ and determine its exact projective degree growth. Its forward and inverse indeterminacy points are distinct, and the exceptional line maps to a forward-fixed point. A recurrence proves $\deg H^n = 2^n$ for every $n \geq 1$, hence algebraic dynamical degree two. We also certify two fixed points and a primitive real two-cycle with exact tangent monodromy; parameters -3 and -5 are exact negative controls for that cycle. These are exact structural data, but there is no complete orbit atlas, prime-like target correspondence, or determinant/analytic bridge.

1 Birational projective geometry

Freeze

$$H(x,y) = (x^2 - 4 - y, x), \qquad H^{-1}(x,y) = (y, y^2 - 4 - x).$$

Direct substitution verifies both inverse compositions. Homogenization gives

$$\bar{H}[X:Y:Z] = [X^2 - 4Z^2 - YZ : XZ : Z^2],$$

$$\bar{H}^{-1}[X:Y:Z] = [YZ : Y^2 - 4Z^2 - XZ : Z^2].$$

The corresponding indeterminacy points are $I_+ = [0:1:0]$ and $I_- = [1:0:0]$. On $Z = 0$ away from I_+, the forward map has image I_-, and $\bar{H}(I_-) = I_-$. Since $\det DH = 1$, the affine map has no exceptional curve; the line at infinity is the only projective exceptional curve. Its orbit is the fixed sequence $I_-, I_-, \ldots$ and therefore never reaches I_+.

2 Degree recurrence

Put $p_{-1} = y$, $p_0 = x$, and

$$p_n = p_{n-1}^2 - 4 - p_{n-2}.$$

Then $H^n = (p_n, p_{n-1})$. Induction shows that p_n has degree 2^n and unique monic leading term x^{2^n}: squaring p_{n-1} supplies that term, whereas the remaining terms have lower degree.

For completeness, set $P_k = Z^{2^k} p_k(X/Z, Y/Z)$ and $d = 2^n$. The projective representative of H^n is

$$[P_n : P_{n-1} Z^{d/2} : Z^d].$$

Its first coordinate contains X^d and is not divisible by Z, while every common factor would have to divide the third coordinate Z^d. The triple therefore has gcd one. Consequently

$$\deg \bar{H}^n = 2^n = (\deg \bar{H})^n, \qquad \lambda_1(\bar{H}) = \lim_{n\to\infty} \deg(H^n)^{1/n} = 2.$$

Thus $\bar{H}$ is algebraically stable at every order, independently of the finite replay cutoff. The exact replay prefix is shown in Table 1.

Table 1: Nonexpanded recursive degree certificate.

n	1	2	3	4	5	6	7	8
$\deg p_n$	2	4	8	16	32	64	128	256
$\deg p_{n-1}$	1	2	4	8	16	32	64	128

3 Exact low-period witnesses

A fixed point has $x = y = q$ and $q^2 - 2q - 4 = 0$, giving

$$(q, q) = (1 + \sqrt{5}, 1 + \sqrt{5}), \quad (1 - \sqrt{5}, 1 - \sqrt{5}).$$

There is also the primitive real cycle

$$(0, -2) \longmapsto (-2, 0) \longmapsto (0, -2).$$

For $B(x) = \left(\begin{smallmatrix} 2x & -1 \\ 1 & 0 \end{smallmatrix}\right)$, its monodromy based at $(0, -2)$ is

$$B(-2)B(0) = \begin{pmatrix} -1 & 4 \\ 0 & -1 \end{pmatrix}, \quad \mathrm{tr} = -2, \quad \det = 1,$$

and $\det(I - zB(-2)B(0)) = (1 + z)^2$.

The witness also fixes the parameter. For $H_c(x, y) = (x^2 + c - y, x)$, each proposed transition has residual $(c + 4, 0)$. Thus $c = -3$ and $c = -5$ give nonzero residuals $(1, 0)$ and $(-1, 0)$, respectively, rather than preserving this cycle.

4 Scope

The evidence stores an exact recurrence DAG, sparse leading certificates, and integer probes through $n = 8$, avoiding full expansion at degree 256. An independent checker reconstructs the complete ledger; a fresh SymPy program passes 97 identities, canonical replay fixes the bytes, and all sixteen hostile edits are rejected. Fixed-date isolated PDF builds agree bytewise and all fonts are embedded.

The all-order degree law, fixed points, and single primitive cycle support only A1 weak: no prime-like orbit-to-target correspondence or complete atlas is present. No weighted dynamical zeta, target divisor, or transfer owner exists, so A2 fails. No functional equation, Gamma-factor treatment, counting law, controlled continuation, or analytic bridge exists, so A3 fails; no natural lift is supplied, so A4 fails. In the repository evaluator vocabulary,

`(A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL), ROUTE_A_EXPLORATORY.`

The algebraic dynamical degree is not promoted to entropy, and we make no claim about arithmetic/local data, Euler factors, root numbers, automorphy, Hilbert–Pólya, Riemann zeros, or Route B. The release scope is the literal

`NO_BAD_EULER_OR_ROOT_NUMBER.`